1.Install Libraries

```
pip install pandas numpy matplotlib seaborn
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/pytho
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/di
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
```

2.Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

3.Load Dataset

```
df = pd.read_csv("titanic.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs	female	38.0	1	0	PC 175

Next steps: Generate code with df New interactive sheet

4.Basic Information

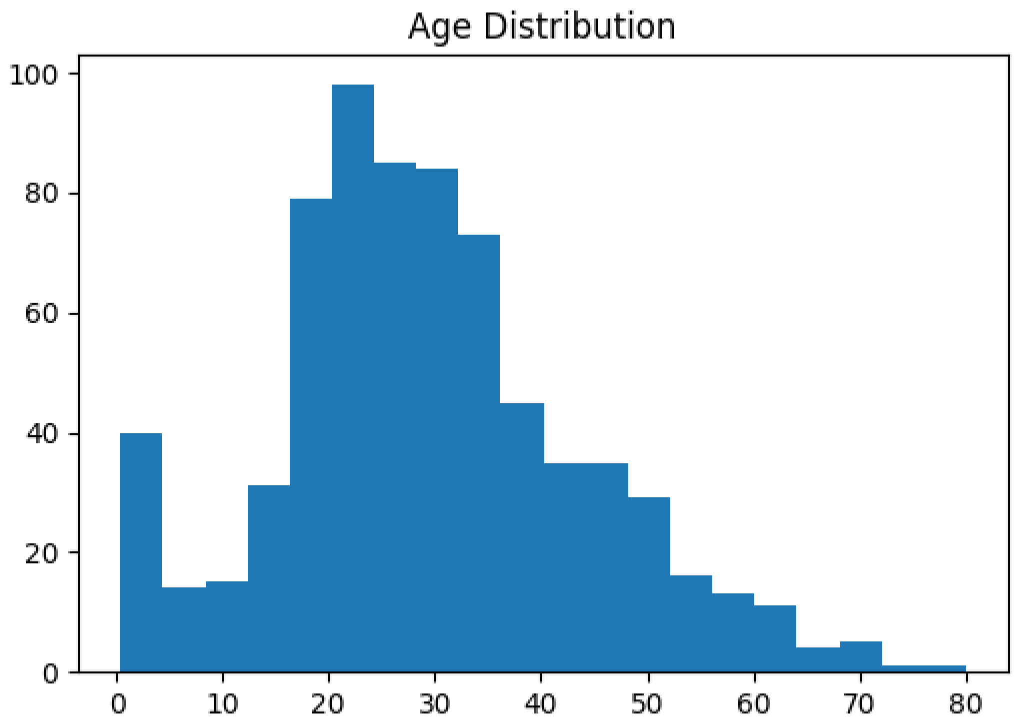
```
# First 5 rows
print(df.head())

# Dataset info
print(df.info())

# Missing values
print(df.isnull().sum())

# Statistics
print(df.describe())
```

```
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
```

```
50%      446.000000    0.000000    3.000000   28.000000    0.000000
75%      668.500000    1.000000    3.000000   38.000000    1.000000
max      891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
```

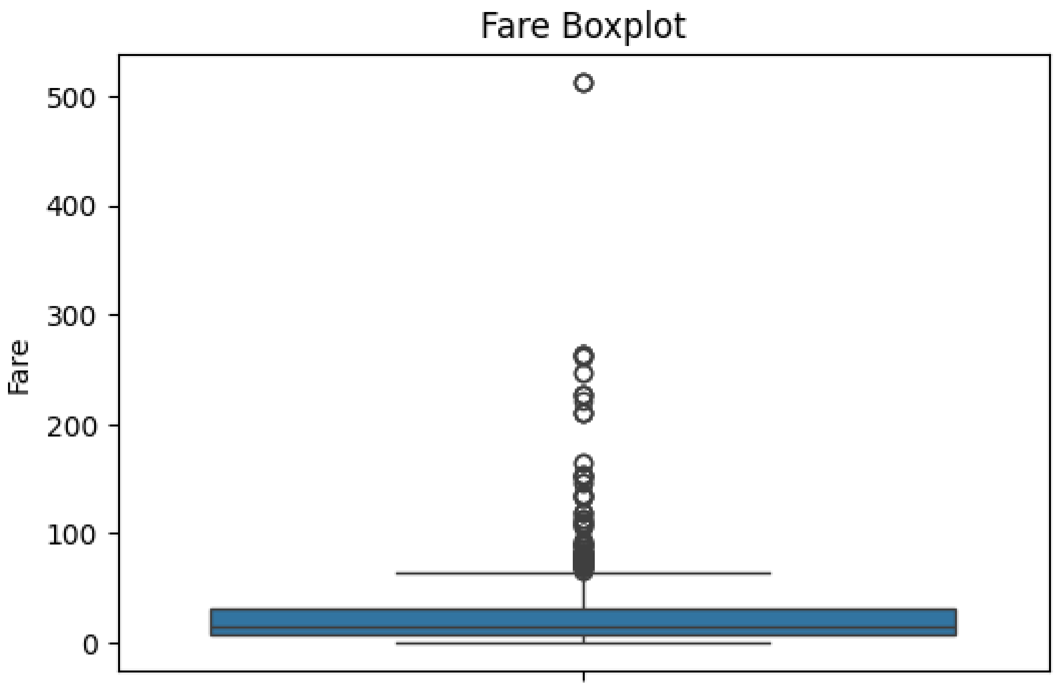
5.Histogram

```
plt.figure(figsize=(6,4))
plt.hist(df['Age'].dropna(), bins=20)
plt.title('Age Distribution')
plt.show()
```



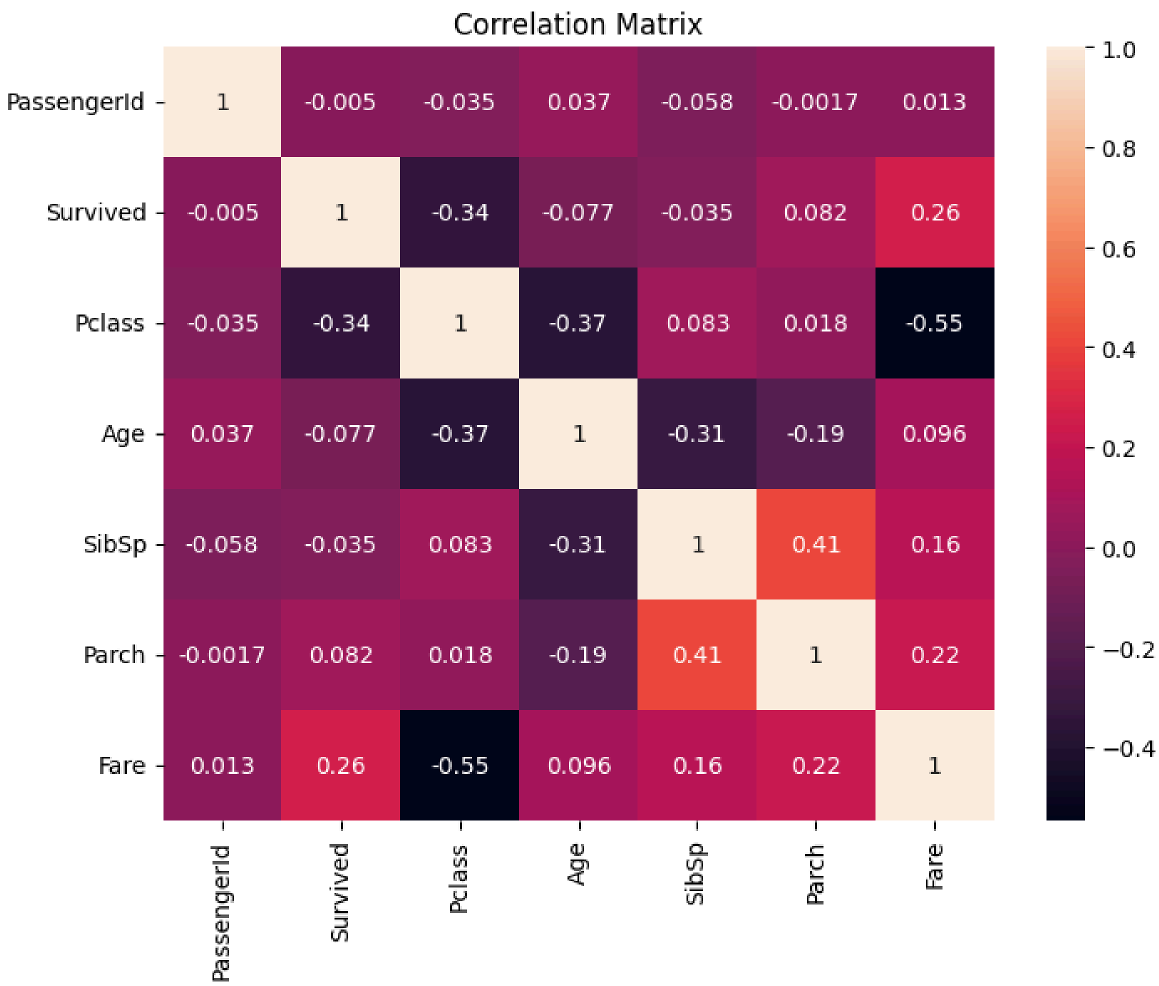
6.Boxplot

```
plt.figure(figsize=(6,4))
sns.boxplot(y=df['Fare'])
plt.title("Fare Boxplot")
plt.show()
```



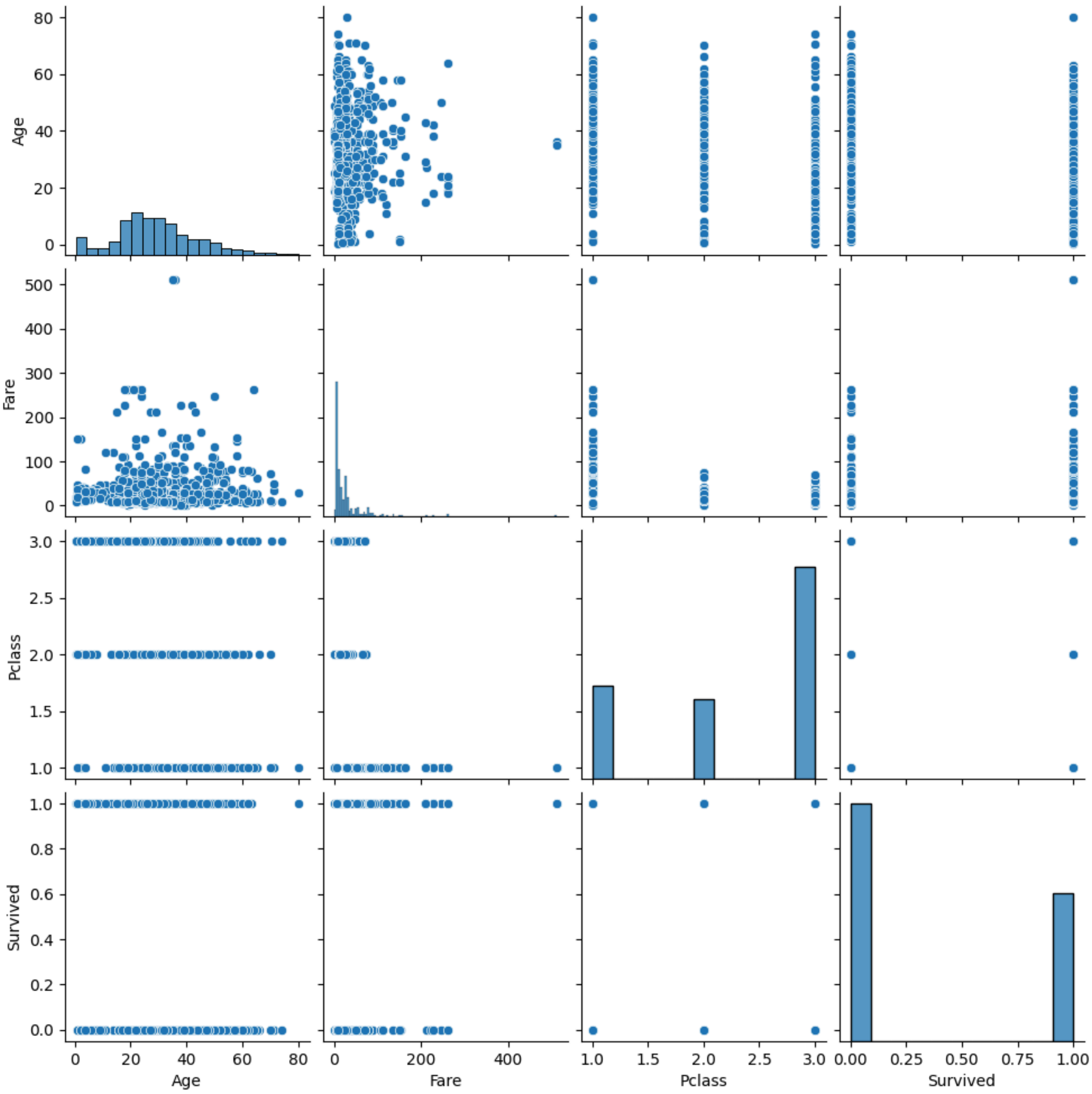
7.Correlation Matrix

```
numeric_df = df.select_dtypes(include='number')

plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True)
plt.title('Correlation Matrix')
plt.show()
```



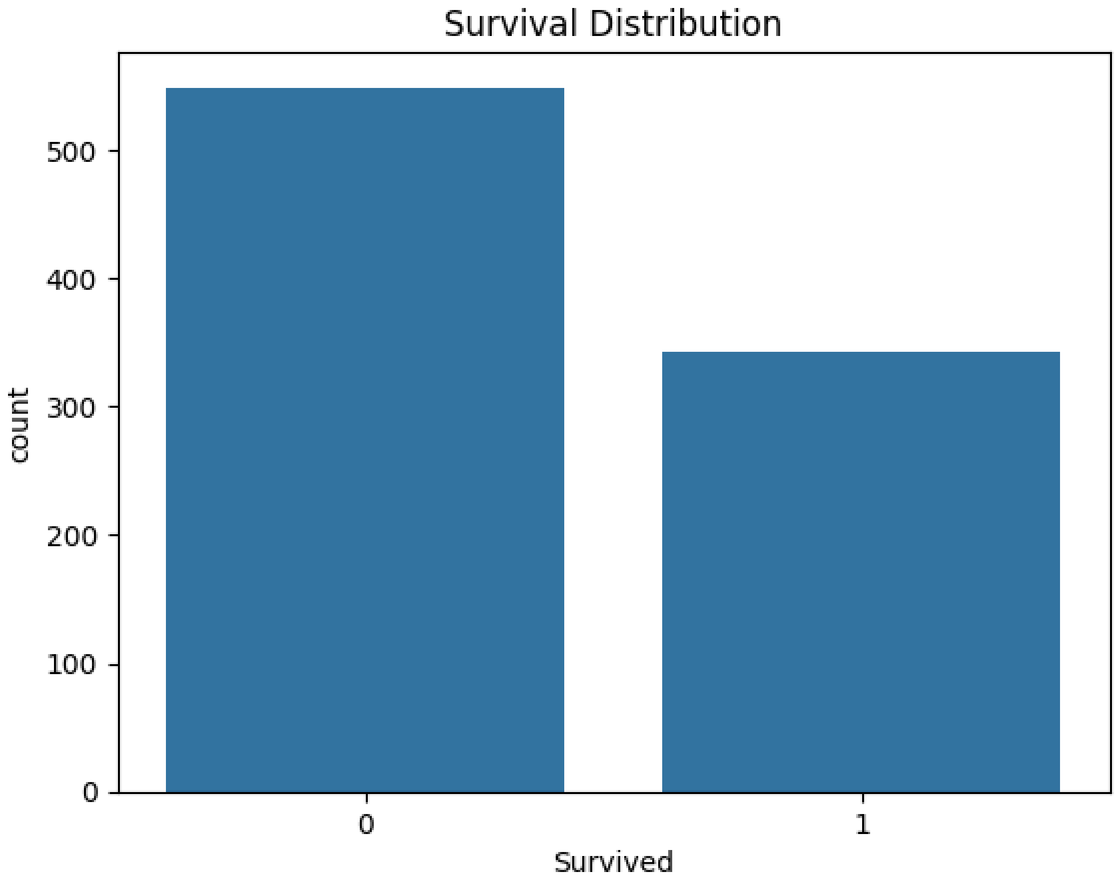
8.Pairplot

```
sns.pairplot(df[['Age','Fare','Pclass','Survived']])
plt.show()
```


Age
Fare
Pclass
Survived
80
60
40
20
0
500
400
300
200
100
0
3.0
2.5
2.0
1.5
1.0
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
Age
0
200
400
Fare
1.0
1.5
2.0
2.5
3.0
Pclass
0.00
0.25
0.50
0.75
1.00
Survived

9 Survival Count Plot

```
sns.countplot(x='Survived', data=df)
plt.title("Survival Distribution")
plt.show()
```



10.Male vs Female Survival

```
sns.countplot(x='Sex', hue='Survived', data=df)
```